



A bi-Stirling-Euler-Mahonian polynomial

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ABSTRACT

Motivated by recent work on (re)mixed Eulerian numbers, we provide a combinatorial interpretation of a subfamily of the remixed Eulerian numbers introduced by Nadeau and Tewari. More specifically, we show that these numbers can be realized as the generating polynomials of permutations with respect to the statistics of left-to-right minima, right-to-left minima, descents, and the mixed major index. Our results generalize both the bi-Stirling-Eulerian polynomials of Carlitz-Scoville and the Stirling-Euler-Mahonian polynomials of Butler.

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1. Introduction

The Eulerian numbers $\langle n \rangle_k$ ($n > k \geq 0$) can be defined by the recurrence

$$\langle n \rangle_k = (n - k) \langle n - 1 \rangle_{k-1} + (k + 1) \langle n - 1 \rangle_k, \quad (1.1)$$

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with initial condition $\langle 0 \rangle = 1$ and $\langle n \rangle = 0$ if $k \notin 0, \dots, n-1$. This recurrence immediately implies the symmetry $\langle n \rangle_k = \langle n \rangle_{n-k-1}$. It is well-known [9,15] that $\langle n \rangle_k$ counts the number of permutations in $\mathfrak{S}_n$ with k descents. The n th Eulerian polynomial, which generates the Eulerian numbers, is defined by

$$A_n(x) = \sum_{k=0}^{n-1} \langle n \rangle_k x^k \quad (1.2)$$

and has the exponential generating function

$$1 + \sum_{n=1}^{\infty} A_n(x) \frac{t^n}{n!} = \frac{1-x}{e^{t(x-1)} - x}. \quad (1.3)$$

Setting $A(r, s) := \langle r+s+1 \rangle_r$, Carlitz-Scoville [7] noticed the following symmetric formula

$$F(x, y) := \sum_{r,s=0}^{\infty} A(r, s) \frac{x^r y^s}{(r+s+1)!} = \frac{e^x - e^y}{xe^y - ye^x}, \quad (1.4)$$

and further introduced the (α, β) -Eulerian numbers $A(r, s | \alpha, \beta)$ by

$$\sum_{r,s=0}^{\infty} A(r, s | \alpha, \beta) \frac{x^r y^s}{(r+s)!} = (1 + xF(x, y))^\alpha (1 + yF(x, y))^\beta. \quad (1.5)$$

Note that $A(r, s | \alpha, \beta) = A(s, r | \beta, \alpha)$ and $A(r, s | 1, 1) = A(r, s)$. The first few terms of the above generating function are

$$1 + (\alpha x + \beta y) + \frac{1}{2}(\alpha^2 x^2 + (2\alpha\beta + \alpha + \beta)xy + \beta^2 y^2) + \frac{1}{6}(\alpha^3 x^3 + \beta^3 y^3 + (3\alpha^2\beta + 3\alpha^2 + 3\alpha\beta + \alpha + \beta)xy^2 + (3\alpha\beta^2 + 3\alpha\beta + 3\beta^2 + \alpha + \beta)xy^2) + \dots \quad (1.6)$$

Carlitz-Scoville [7, §8] proved the recurrence

$$A(r, s | \alpha, \beta) = (s + \alpha)A(r-1, s | \alpha, \beta) + (r + \beta)A(r, s-1 | \alpha, \beta) \quad (1.7)$$

and also provided the combinatorial interpretation

$$A(r, s | \alpha, \beta) = \sum_{\substack{\sigma \in \mathfrak{S}_{r+s+1} \\ \text{des}(\sigma)=r}} \alpha^{\text{lrmin}(\sigma)-1} \beta^{\text{rlmin}(\sigma)-1}, \quad (1.8)$$

where $\text{des}(\sigma)$ is the number of descents of σ , while $\text{lrmin}(\sigma)$ (resp. $\text{rlmin}(\sigma)$) is the number of *left-to-right* (resp. *right-to-left*) minima of σ , see (1.11).

Let $\lambda_1 \geq \dots \geq \lambda_{r+1}$ be real numbers and let $\lambda = (\lambda_1, \dots, \lambda_{r+1})$. The permutahedron $\text{Perm}(\lambda)$ is the convex hull of points $\lambda_\sigma = (\lambda_{\sigma(1)}, \dots, \lambda_{\sigma(r+1)})$ where σ is a permutation in $\mathfrak{S}_{r+1}$. It is known that

$$\text{vol}(\text{Perm}(\lambda)) = \sum_{c=(c_1, \dots, c_r)} A_c \frac{\mu_1^{c_1} \dots \mu_r^{c_r}}{c_1! \dots c_r!}$$

where $c \in \mathcal{W}_r = \{(c_1, \dots, c_r) : c_1 + \dots + c_r = r\}$ and $\mu_i = \lambda_i - \lambda_{i+1}$. The coefficients A_c are positive integers known as the *mixed Eulerian numbers* [16].

In this paper we adopt the standard q -notations [1]. For $x \in \mathbb{R}$ the q -real number $[x]$ is defined by

$$[x] = \frac{1 - q^x}{1 - q}.$$

For an integer $r \in \mathbb{N}$, we define $[r]! = 1 \cdot (1 + q) \cdots (1 + q + \dots + q^{r-1})$ and the q -binomial coefficient by

$$\begin{bmatrix} r \\ k \end{bmatrix} = \frac{[r]!}{[k]![r-k]!} \quad \text{for } r \geq k \geq 0.$$

Nadeau and Tewari [13,14] introduced a q -analogue of the mixed Eulerian numbers $A_c(q)$, known as the *remixed Eulerian numbers*. These numbers possess several remarkable properties, among which:

- (1) $A_c(q)$ is a polynomial with non-negative coefficients and $A_c(1) = A_c$,
- (2) $A_{(c_1, \dots, c_r)}(q) = q^{\binom{r}{2}} A_{(c_r, \dots, c_1)}(q^{-1})$,
- (3) $\sum_{c \in \mathcal{W}_r} A_c(q) = [r]! \cdot \frac{1}{r+1} \binom{2r}{r}$,
- (4) $A_{(1, \dots, 1)}(q) = [r]!$, and
- (5) $A_{(k, 0, \dots, 0, r-k)}(q) = q^{\binom{k}{2}} \cdot \begin{bmatrix} r \\ k \end{bmatrix}$.

Recently, Gaudin [11] observed that the Carlitz-Scoville numbers $A(r, s | \alpha, \beta)$ are a rescaled subfamily of the mixed Eulerian numbers and proposed the following q -analogue $A(r, s | \alpha, \beta)_q$ within the framework of the remixed Eulerian numbers,

$$A(r, s | \alpha, \beta)_q = \frac{1}{[\alpha + \beta - 1]!} A_{(0^r, 1^{\alpha-1}, r+s+1, 1^{\beta-1}, 0^s)}(q). \quad (1.9)$$

The following q -analogue of the recurrence (1.7) is due to Gaudin [11, Proposition 4.3]:

$$A(r, s | \alpha, \beta)_q = q^{r+\beta-1} [s + \alpha] A(r-1, s | \alpha, \beta)_q + [r + \beta] A(r, s-1 | \alpha, \beta)_q. \quad (1.10)$$

For $\sigma = \sigma_1 \sigma_2 \dots \sigma_n \in \mathfrak{S}_n$, define the following statistics

$$\text{des}(\sigma) = |\{i \in [n-1] \mid \sigma_i > \sigma_{i+1}\}|, \quad (1.11a)$$

$$\text{lrmin}(\sigma) = |\{i \in [n] \mid \sigma_i < \sigma_j, \forall j < i\}|, \quad (1.11b)$$

$$\text{rlmin}(\sigma) = |\{i \in [n] \mid \sigma_i < \sigma_j, \forall j > i\}|. \quad (1.11c)$$

The statistic des is a well-known Eulerian statistic due to (1.2), both lrmin and rlmin are Stirling statistics:

$$\sum_{\sigma \in \mathfrak{S}_n} x^{\text{lrmin}(\sigma)} = \sum_{\sigma \in \mathfrak{S}_n} x^{\text{rlmin}(\sigma)} = \sum_{k=0}^n s(n, k) x^k, \quad (1.12)$$

where $s(n, k)$ are the Stirling numbers of the first kind. The *inversion number* inv and *major index* maj are defined by

$$\text{inv}(\sigma) = |\{(i, j) \mid \sigma_i > \sigma_j, i < j\}|, \quad (1.13)$$

$$\text{maj}(\sigma) = \sum_{\sigma_i > \sigma_{i+1}} i. \quad (1.14)$$

The statistics inv and maj are well-known Mahonian statistics:

$$\sum_{\sigma \in \mathfrak{S}_n} q^{\text{inv}(\sigma)} = \sum_{\sigma \in \mathfrak{S}_n} q^{\text{maj}(\sigma)} = [n]!. \quad (1.15)$$

For convenience, we introduce the following notation, valid for $n \geq k \geq 0$:

$$E_{n,k}(\alpha, \beta, q) := A(k, n-k \mid \alpha, \beta)_q \quad (1.16)$$

with the convention $E_{n,k}(\alpha, \beta, q) = 0$ if $k \notin \{0, \dots, n\}$. As $A_{(1^r)}(q) = [r]!$ (see [14]), by (1.9) we have $E_{0,0}(\alpha, \beta, q) = 1$. In view of (1.16), setting $r+s=n$ and $r=k$ in (1.10) we obtain

$$E_{n,k}(\alpha, \beta, q) = q^{\beta+k-1} [n-k+\alpha] E_{n-1,k-1}(\alpha, \beta, q) + [k+\beta] E_{n-1,k}(\alpha, \beta, q). \quad (1.17)$$

When $(\alpha, \beta) = (1, 1)$ or $(0, 1)$, several q -analogues of Eulerian numbers are already known [6, 18, 17]. In the case $\alpha = 0$, Butler [4, 5] introduced a novel q -analogue of Eulerian numbers by combining the parameter β with the major index.

The aim of this paper is to provide a combinatorial interpretation of $E_{n,k}(\alpha, \beta, q)$ in terms of the permutation statistics such as left-to-right minima, right-to-left minima, descents and (re)mixed major index, and to derive several explicit formulas for these numbers. Our results generalize and unify those of Carlitz-Scoville [7] and Butler [4].

For $\sigma \in \mathfrak{S}_n$ the *mixed major index* of σ , denoted by $\widetilde{\text{maj}}$, is defined by

$$\widetilde{\text{maj}}(\sigma) = (1 + \text{des}(\sigma) - \text{lrmin}(\sigma))(\alpha - 1) + (n - \text{rlmin}(\sigma))(\beta - 1) + \text{maj}(\sigma). \quad (1.18)$$

Theorem 1.1. For integer $n \geq k \geq 0$, we have

$$E_{n,k}(\alpha, \beta, q) = \sum_{\sigma \in \mathfrak{S}_{n+1,k}} [\alpha]^{\text{lrmin}(\sigma)-1} [\beta]^{\text{rlmin}(\sigma)-1} q^{\widetilde{\text{maj}}(\sigma)}, \quad (1.19)$$

where $\mathfrak{S}_{n+1,k}$ denotes the set of permutations in $\mathfrak{S}_{n+1}$ with k descents.

We define the *bi-Stirling-Euler-Mahonian polynomials* $E_n(x \mid \alpha, \beta, q)$ by

$$E_n(x \mid \alpha, \beta, q) = \sum_{k=0}^n E_{n,k}(\alpha, \beta, q) x^k. \quad (1.20)$$

For example, it follows from (1.17) that

$$\begin{aligned} E_0(x \mid \alpha, \beta, q) &= 1, \\ E_1(x \mid \alpha, \beta, q) &= [\beta] + q^\beta [\alpha] x, \\ E_2(x \mid \alpha, \beta, q) &= [\beta]^2 + q^\beta ([1 + \alpha][\beta] + [\alpha][\beta + 1])x + q^{2\beta+1} [\alpha]^2 x^2. \end{aligned}$$

Define the q -derivative operator δ_x by

$$\delta_x p(x) = \frac{p(qx) - p(x)}{(q-1)x}, \quad (1.21)$$

where $p(x) \in \mathbb{R}[[x]]$.

Theorem 1.2. Let $E_n(x; q) := E_n(x \mid \alpha, \beta, q)$ for $n \geq 0$. We have the following q -identities:

(i)

$$E_n(x; q) = ([\beta] + q^\beta [n-1 + \alpha]x) E_{n-1}(x; q) + (1-x)xq^\beta \delta_x(E_{n-1}(x; q)). \quad (1.22)$$

(ii)

$$\frac{E_n(x; q)}{(x; q)_{n+\alpha+\beta}} = \sum_{j \geq 0} x^j \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} [j + \beta]^n. \quad (1.23)$$

(iii)

$$\sum_{n \geq 0} \frac{E_n(x; q)}{(x; q)_{n+\alpha+\beta}} \frac{t^n}{n!} = \sum_{j \geq 0} x^j \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} \exp([j + \beta]t). \quad (1.24)$$

(iv)

$$\sum_{k=0}^j \begin{bmatrix} j - k + n + \alpha + \beta - 1 \\ j - k \end{bmatrix} E_{n,k}(\alpha, \beta, q) = \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} [j + \beta]^n. \quad (1.25)$$

(v)

$$E_{n,k}(\alpha, \beta, q) = \sum_{j=0}^k \begin{bmatrix} n + \alpha + \beta \\ k - j \end{bmatrix} \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} (-1)^{k-j} q^{\binom{k-j}{2}} [\beta + j]^n. \quad (1.26)$$

We shall prove Theorem 1.1 and Theorem 1.2 in Section 2 and derive some corollaries in Section 3. Barbero et al. [2] introduced the ν -order (s, t) -Eulerian numbers. In Section 4 we show that when $\nu = 1$ their numbers coincide with the Carlitz-Scoville numbers $A(r, s | \alpha, \beta)$.

Remark 1. Using the inversion statistic on permutations, Dong-Lin-Pan [8] obtained a q -analogue of (1.5) via Gessel's q -compositional formula. For recent developments on Carlitz-Scoville's generalized Eulerian polynomials, we refer the reader to [19].

2. Proof of main theorems

Proof of Theorem 1.1. Consider the enumerative polynomials

$$\widehat{E}_{n,k}(\alpha, \beta, q) = \sum_{\sigma \in \mathfrak{S}_{n+1,k}} [\alpha]^{\text{lrmin}(\sigma)-1} [\beta]^{\text{rlmin}(\sigma)-1} q^{\widetilde{\text{maj}}(\sigma)}. \quad (2.1)$$

It suffices to verify that the polynomials $\widehat{E}_{n,k}(\alpha, \beta, q)$ satisfy the recurrence (1.17). Clearly Eq. (1.17) is valid for $n = 0$. It is clear that deleting $n + 1$ from a permutation in $\mathfrak{S}_{n+1,k}$, we obtain a permutation in $\mathfrak{S}_{n,k} \cup \mathfrak{S}_{n,k-1}$. Inversely, any permutation π in $\mathfrak{S}_{n+1,k}$ can be created by adding $n + 1$ to a permutation w in $\mathfrak{S}_{n,k} \cup \mathfrak{S}_{n,k-1}$. We will discuss four types of insertion positions: a) at the beginning of w , b) at the end of w , c) at descent positions of w , d) at ascent positions of w .

(a) If $w = \sigma_1 \dots \sigma_n \in \mathfrak{S}_{n,k-1}$, then $\sigma = (n + 1)\sigma_1 \dots \sigma_n \in \mathfrak{S}_{n+1,k}$ with

$$(\text{lrmin}, \text{rlmin}, \text{des})\pi = (1 + \text{lrmin}, \text{rlmin}, 1 + \text{des})w.$$

Hence

$$\sum_{\substack{\pi \in \mathfrak{S}_{n+1,k} \\ \pi(1)=n+1}} [\alpha]^{\text{lrmin}(\pi)-1} [\beta]^{\text{rlmin}(\pi)-1} q^{\widetilde{\text{maj}}(\pi)} = [\alpha]q^{\beta+k-1} \widehat{E}_{n-1,k-1}(\alpha, \beta, q).$$

(b) If $w = \sigma_1 \dots \sigma_n \in \mathfrak{S}_{n,k}$, then $\pi = \sigma_1 \dots \sigma_n(n + 1) \in \mathfrak{S}_{n+1,k}$ and $(\text{lrmin}, \text{rlmin})\pi = (\text{lrmin}, 1 + \text{rlmin})w$. Hence

$$\sum_{\substack{\pi \in \mathfrak{S}_{n+1,k} \\ \pi(n)=n+1}} [\alpha]^{\text{lrmin}(\pi)-1} [\beta]^{\text{rlmin}(\pi)-1} q^{\widetilde{\text{maj}}(\pi)} = [\beta] \widehat{E}_{n-1,k}(\alpha, \beta, q).$$

(c) If $w = \sigma_1 \dots \sigma_n \in \mathfrak{S}_{n,k}$ with the descent set $\{i_1, i_2, \dots, i_k\}$, then we define $\pi = \sigma_1 \dots \sigma_{i_j}(n + 1)\sigma_{i_j+1} \dots \sigma_n \in \mathfrak{S}_{n+1,k}$ with $1 \leq j \leq k$. It is clear that $(\text{lrmin}, \text{rlmin})\pi = (\text{lrmin}, \text{rlmin})w$ and $\widetilde{\text{maj}}(\pi) = \widetilde{\text{maj}}(w) + k - j$. Thus

$$\sum_{\pi \in \mathfrak{S}'_{n+1,k}} [\alpha]^{\text{lrmin}(\pi)-1} [\beta]^{\text{rlmin}(\pi)-1} q^{\widetilde{\text{maj}}(\pi)} = q^\beta [k] \widehat{E}_{n-1,k}(\alpha, \beta, q),$$

where $\mathfrak{S}'_{n+1,k}$ is the set of permutations $\pi = \sigma_1 \dots \sigma_i(n+1)\sigma_{i+1} \dots \sigma_n$ in $\mathfrak{S}_{n+1,k}$ such that $\sigma_i > \sigma_{i+1}$.

- (d) Let $w = \sigma_1 \dots \sigma_n \in \mathfrak{S}_{n,k-1}$ with descent set $\{i_1, \dots, i_{k-1}\}$. If π is a permutation $\pi = \sigma_1 \dots \sigma_j(n+1)\sigma_{j+1} \dots \sigma_n \in \mathfrak{S}_{n+1,k}$ with $i_l < j < i_{l+1}$, where $0 \leq l \leq k-1$, $i_0 = 0$ and $i_k = n$, then

$$(\text{lrmin}, \text{rlmin}, \text{des})\pi = (\text{lrmin}, \text{rlmin}, 1 + \text{des})w, \quad (2.2)$$

$$\widetilde{\text{maj}}(\pi) = (k-1-l) + (j+1) + (\alpha-1) + (\beta-1) + \widetilde{\text{maj}}(w). \quad (2.3)$$

Since

$$\begin{aligned} \bigcup_{l=0}^{k-1} \{k-1-l+j : i_l < j < i_{l+1}\} &= \bigcup_{l=0}^{k-1} \{k-l+i_l, \dots, k-l+i_{l+1}-2\} \\ &= \{k, k+1, \dots, n-1\}, \end{aligned}$$

we have

$$\sum_{\substack{i_l < j < i_{l+1} \\ 0 \leq l \leq k-1}} q^{\alpha+\beta-1+k-1-l+j} = q^{\alpha+\beta+k-1} [n-k].$$

It follows that

$$\sum_{\pi \in \mathfrak{S}''_{n+1,k}} [\alpha]^{\text{lrmin}(\pi)-1} [\beta]^{\text{rlmin}(\pi)-1} q^{\widetilde{\text{maj}}(\pi)} = q^{\alpha+\beta+k-1} [n-k] \cdot \widehat{E}_{n-1,k-1}(\alpha, \beta, q),$$

where $\mathfrak{S}''_{n+1,k}$ is the set of permutations $\sigma = \sigma_1 \dots \sigma_i(n+1)\sigma_{i+1} \dots \sigma_n$ in $\mathfrak{S}_{n+1,k}$ such that $\sigma_i < \sigma_{i+1}$.

Combining the two cases (a) and (d) (resp. (b) and (c)) we obtain

$$\widehat{E}_{n,k}(\alpha, \beta, q) = q^{\beta+k-1} [n-k+\alpha] \widehat{E}_{n-1,k-1}(\alpha, \beta, q) + [k+\beta] \widehat{E}_{n-1,k}(\alpha, \beta, q). \quad (2.4)$$

Comparing the above equation with (1.17) we obtain $\widehat{E}_{n,k}(\alpha, \beta, q) = E_{n,k}(\alpha, \beta, q)$. $\square$

For non-negative integer $n \in \mathbb{Z}$, we define $(x; q)_0 = 1$ and

$$(x; q)_n = (1-x)(1-xq) \cdots (1-xq^{n-1}) \quad (n \geq 1).$$

The q -binomial coefficients $\begin{bmatrix} x \\ n \end{bmatrix}$ are defined by

$$\begin{bmatrix} x \\ n \end{bmatrix} = \frac{(q^{x-n+1}; q)_n}{(q; q)_n}.$$

Furthermore we define $(x; q)_\infty = \lim_{n \rightarrow \infty} (x; q)_n$ and $(x; q)_n$ for all real numbers n by

$$(x; q)_n = (x; q)_\infty / (xq^n; q)_\infty. \quad (2.5)$$

Recall [1, Chapter 2] that

$$\sum_{n=0}^{\infty} \frac{(x; q)_n t^n}{(q; q)_n} = \frac{(xt; q)_\infty}{(t; q)_\infty}. \quad (2.6)$$

In particular we have

$$(x; q)_N = \sum_{i=0}^{\infty} \begin{bmatrix} N \\ i \end{bmatrix} (-1)^i x^i q^{i(i-1)/2}, \quad (2.7)$$

$$\frac{1}{(x; q)_N} = \sum_{j=0}^{\infty} x^j \begin{bmatrix} N+j-1 \\ j \end{bmatrix}. \quad (2.8)$$

For $k, n \in \mathbb{N}$ an easy calculation yields

$$\delta_x \frac{x^k}{(x; q)_n} = x^{k-1} \frac{[k] + x[n-k]q^k}{(x; q)_{n+1}} = \begin{cases} [k]x^{k-1}, & \text{if } n = 0; \\ \frac{[n]}{(x; q)_{n+1}}, & \text{if } k = 0. \end{cases} \quad (2.9)$$

Proof of Theorem 1.2. (i) Multiplying Eq. (1.17) or (2.4) by x^k and summing k from 0 to n we obtain

$$\begin{aligned} & \sum_{k=0}^n E_{n,k}(\alpha, \beta, q) x^k \\ &= \sum_{k=0}^n \{q^{\beta+k-1} [n-k+\alpha] E_{n-1,k-1}(\alpha, \beta, q) + [k+\beta] E_{n-1,k}(\alpha, \beta, q)\} x^k \\ &= \sum_{k=0}^{n-1} q^{\beta+k} [n-1+\alpha-k] E_{n-1,k}(\alpha, \beta, q) x^{k+1} + \sum_{k=0}^{n-1} [k+\beta] E_{n-1,k}(\alpha, \beta, q) x^k. \end{aligned} \quad (2.10)$$

Substituting $[k+\beta] = [\beta] + q^\beta [k]$ and $q^{\beta+k} [n-1+\alpha-k] = q^\beta [n-1+\alpha] - q^\beta [k]$ in (2.10) and then using $\delta_x x^k = [k]x^{k-1}$, we have

$$\begin{aligned} E_n(x; q) &= q^\beta [n-1+\alpha] x E_{n-1}(x; q) - q^\beta x^2 \delta_x (E_{n-1}(x; q)) \\ &\quad + [\beta] E_{n-1}(x; q) + q^\beta x \delta_x (E_{n-1}(x; q)), \end{aligned} \quad (2.11)$$

which is clearly equal to Eq. (1.22).

(ii) We proceed by induction on $n \geq 0$. When $n = 0$ Eq. (1.23) is (2.8) with $N = \alpha + \beta$. Multiplying (1.23) by x^β and shifting $n \rightarrow n - 1$ yields

$$\frac{x^\beta E_{n-1}(x; q)}{(x; q)_{n+\alpha+\beta-1}} = \sum_{j \geq 0} x^{j+\beta} \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} [j + \beta]^{n-1}. \quad (2.12)$$

Applying the operator $x^{1-\beta} \delta_x$ to (2.12) and using the q -Leibniz formula

$$\delta_x(f(x)g(x)) = \delta_x(f(x))g(qx) + f(x)\delta_x(g(x)), \quad (2.13)$$

we obtain

$$\begin{aligned} & \frac{(1-x)(q^\beta x \delta_x(E_n(x; q)) + [\beta]E_n(x; q)) + [n + \alpha + \beta - 1]xE_n(x; q)}{(x; q)_{n+\alpha+\beta}} \\ &= \sum_{j \geq 0} x^j \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} [j + \beta]^n. \end{aligned} \quad (2.14)$$

Combining the above identity with (1.22), we have (1.23).

(iii) Multiplying Eq. (1.23) by $t^n/n!$ followed by summing over $n \geq 0$, we have

$$\begin{aligned} \sum_{n \geq 0} \frac{E_n(x; q)}{(x; q)_{n+\alpha+\beta}} \frac{t^n}{n!} &= \sum_{j \geq 0} x^j \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} \sum_{n \geq 0} \frac{([j + \beta]t)^n}{n!} \\ &= \sum_{j \geq 0} x^j \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} \exp([j + \beta]t). \end{aligned} \quad (2.15)$$

(iv) Eq. (1.25) follows by extracting the coefficient of x^j in both sides of Eq. (1.23) and applying Eq. (2.8).

(v) Eq. (1.26) follows from multiplying both sides of Eq. (1.23) by $(x; q)_{n+\alpha+\beta}$ and extracting the coefficient of x^k in both sides by using Eq. (2.7). $\square$

3. Corollaries of main theorems

In this section, we illustrate applications of our main theorem by specializing its parameters.

Corollary 3.1. [3, Corollary 4.2.4] For $n \geq 1$, we have

$$\frac{\sum_{\sigma \in \mathfrak{S}_n} x^{\text{des}(\sigma)} q^{\widetilde{\text{maj}}_1(\sigma)} [\beta]^{\text{rlmin}(\sigma)}}{(x; q)_{n+\beta}} = \sum_{j \geq 0} x^j \begin{bmatrix} j + \beta - 1 \\ j \end{bmatrix} [j + \beta]^n, \quad (3.1)$$

where $\widetilde{\text{maj}}_1(\sigma) = (n - \text{rlmin}(\sigma))(\beta - 1) + \text{maj}(\sigma)$.

Proof. When $\alpha = 0$, the summation for $E_n(x | \alpha, \beta, q)$ reduces to the permutations $\sigma \in \mathfrak{S}_{n+1}$ such that $\text{lrmin}(\sigma) = 1$, viz, $\sigma_1 = 1$. Define $\sigma'_i = \sigma_{i+1} - 1$ for $i \in [n]$, then $\sigma' \in \mathfrak{S}_n$ and

- $\text{des}(\sigma) = \text{des}(\sigma')$;
- $\text{maj}(\sigma) = \text{maj}(\sigma') + \text{des}(\sigma')$;
- $\text{rlmin}(\sigma) = \text{rlmin}(\sigma') + 1$.

Hence,

$$\sum_{\substack{\sigma \in \mathfrak{S}_{n+1} \\ \text{lrmin}(\sigma)=1}} x^{\text{des}(\sigma)} q^{\widetilde{\text{maj}}(\sigma)} [\beta]^{\text{rlmin}(\sigma)-1} = \sum_{\sigma \in \mathfrak{S}_n} x^{\text{des}(\sigma)} q^{\widetilde{\text{maj}}_1(\sigma)} [\beta]^{\text{rlmin}(\sigma)}. \quad (3.2)$$

The result follows by combining (1.23) and (3.2). $\square$

Corollary 3.2. For $n \geq 1$, we have

$$\frac{\sum_{\sigma \in \mathfrak{S}_n} x^{\text{des}(\sigma)} q^{\widetilde{\text{maj}}_2(\sigma)} [\alpha]^{\text{lrmin}(\sigma)}}{(x; q)_{n+\alpha}} = \sum_{j \geq 0} x^j \begin{bmatrix} j + \alpha \\ j + 1 \end{bmatrix} [j + 1]^n, \quad (3.3)$$

where $\widetilde{\text{maj}}_2(\sigma) = (1 + \text{des}(\sigma) - \text{lrmin}(\sigma))(\alpha - 1) + \text{maj}(\sigma)$.

Proof. When $\beta = 0$, the summation for $E_n(x | \alpha, \beta, q)$ reduces to the permutations $\sigma \in \mathfrak{S}_{n+1}$ with $\sigma_{n+1} = 1$. Let $\sigma'_i = \sigma_i - 1$ for $i \in [n]$, then $\sigma' \in \mathfrak{S}_n$ and

- $\text{des}(\sigma) = \text{des}(\sigma') + 1$;
- $\text{maj}(\sigma) = \text{maj}(\sigma') + n$;
- $\text{lrmin}(\sigma) = \text{lrmin}(\sigma') + 1$.

Hence,

$$\sum_{\substack{\sigma \in \mathfrak{S}_{n+1} \\ \text{rlmin}(\sigma)=1}} x^{\text{des}(\sigma)} q^{\widetilde{\text{maj}}(\sigma)} [\alpha]^{\text{lrmin}(\sigma)-1} = \sum_{\sigma \in \mathfrak{S}_n} x^{1+\text{des}(\sigma)} q^{\widetilde{\text{maj}}_2(\sigma)} [\alpha]^{\text{lrmin}(\sigma)}. \quad (3.4)$$

The result follows by combining (1.23) and (3.4). $\square$

Setting $\alpha = 1$ in (3.3) or $\beta = 1$ in (3.1) we recover Carlitz's identity [6],

$$\frac{\sum_{\sigma \in \mathfrak{S}_n} x^{\text{des}(\sigma)} q^{\text{maj}(\sigma)}}{(x; q)_{n+1}} = \sum_{j \geq 0} x^j [j + 1]^n. \quad (3.5)$$

Corollary 3.3. For $n \geq 1$ we have

$$E_n(1; q) = \prod_{i=0}^{n-1} [\alpha + \beta + i]. \quad (3.6)$$

Proof. Note that

$$\frac{E_n(1, q)}{(q; q)_{n+\alpha+\beta-1}} = \lim_{x \rightarrow 1^-} \frac{E_n(x, q)}{(xq; q)_{n+\alpha+\beta-1}}.$$

Combining (1.23) and Abel's lemma we have

$$\begin{aligned} E_n(1, q) &= (q; q)_{n+\alpha+\beta-1} \lim_{x \rightarrow 1^-} (1-x) \sum_{j \geq 0} x^j \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} [j + \beta]^n \\ &= (q; q)_{n+\alpha+\beta-1} \lim_{j \rightarrow +\infty} \begin{bmatrix} j + \alpha + \beta - 1 \\ j \end{bmatrix} [j + \beta]^n \\ &= \frac{(q; q)_{n+\alpha+\beta-1}}{(q; q)_{\alpha+\beta-1} (1-q)^n}, \end{aligned}$$

which is equivalent to (3.6). $\square$

Remark 2. Alternatively, setting $x = 1$ in (1.22) yields

$$E_n(1; q) = [n - 1 + \alpha + \beta] E_{n-1}(1; q).$$

As $E_0(1; q) = 1$, we obtain (3.6) by iteration. Moreover, we derive from (1.17) that

$$E_{n,0}(\alpha, \beta, q) = [\beta]^n, \quad E_{n,n}(\alpha, \beta, q) = q^{n\beta + \binom{n}{2}} [\alpha]^n. \quad (3.7)$$

Let $\sigma = \sigma_1 \sigma_2 \dots \sigma_n \in \mathfrak{S}_n$. For an integer $r \geq 1$, as a generalization of descents, the set of r -descents of σ is defined by, see [9],

$$r\text{Des}(\sigma) := \{i \in [n-1] : \sigma_i \geq \sigma_{i+1} + r\}. \quad (3.8)$$

The cardinality of $r\text{Des}(\sigma)$ is denoted by $r\text{des}(\sigma)$.

Rawlings [17] introduced the r -major index as

$$r\text{maj}(\sigma) := \sum_{i \in r\text{Des}(\sigma)} i + |r\text{Inv}(\sigma)|, \quad (3.9)$$

where

$$r\text{Inv}(\sigma) := \{(i, j) : 1 \leq i < j \leq n, \sigma_i > \sigma_j > \sigma_i - r\}.$$

He then defined the (q, r) -Eulerian numbers

$$A[n, k; r] := \sum_{\substack{\sigma \in \mathfrak{S}_n \\ \text{rdes}(\sigma) = k}} q^{\text{rmaj}(\sigma)}. \quad (3.10)$$

Proposition 3.4. For $r \geq 1$ and $0 \leq k \leq n$, we have

$$A[n + r, k; r] = [r]! \cdot E_{n,k}(1, r, q). \quad (3.11)$$

Proof. It is known [17] that the (q, r) -Eulerian numbers satisfy the recurrence

$$A[n, k; r] = [k + r]A[n - 1, k; r] + q^{k+r-1}[n + 1 - k - r]A[n - 1, k - 1; r], \quad (3.12)$$

for $n \geq 1$, $0 \leq k \leq n - r$ and $A[r, 0; r] = [r]!$. Now, shifting $n \rightarrow n + r$ Eq. (3.12) becomes

$$A[n + r, k; r] = [k + r]A[n + r - 1, k; r] + q^{k+r-1}[n + 1 - k]A[n + r - 1, k - 1; r] \quad (3.13)$$

for $0 \leq k \leq n$. Comparing (3.13) with recurrence (1.17) and their initial values at $(n, k) = (0, 0)$, we derive (3.11). $\square$

Combining Eq. (1.23) and (3.11), we derive

$$\frac{\sum_{0 \leq k \leq n} A[n + r, k; r] x^k}{(x; q)_{n+1+r}} = \sum_{j \geq 0} x^j \frac{[j + r]!}{[j]!} [j + r]^n. \quad (3.14)$$

From the above equality, we see that $A[n + r, k; r]$ for $0 \leq k \leq n$ can be seen as q -hit numbers of Garsia-Remmel [10].

4. Concluding remarks

In (1.17), let us define

$$E_{n,k}(\alpha, \beta, q) = q^{k\beta + k(k-1)/2} \cdot E_{n,k}^*(\alpha, \beta, q). \quad (4.1)$$

Then

$$E_{n,k}^*(\alpha, \beta, q) = [n - k + \alpha]E_{n-1,k-1}^*(\alpha, \beta, q) + [k + \beta]E_{n-1,k}^*(\alpha, \beta, q). \quad (4.2)$$

It follows that

$$E_{n,k}^*(\alpha, \beta, q) = E_{n,n-k}^*(\beta, \alpha, q) \quad (0 \leq k \leq n). \quad (4.3)$$

Barbero et al. [2] introduced a three-parameter generalization of the standard Eulerian numbers $\langle n \rangle_{(s,t)}^{(\nu)}$ for integer $\nu \geq 1$, called the ν -order (s,t) -Eulerian numbers. When $\nu = 1$, the numbers $\langle n \rangle_{(s,t)} := \langle n \rangle_{(s,t)}^{(1)}$ satisfy the following recurrence

$$\langle n \rangle_{(s,t)} = (k+s) \langle n-1 \rangle_{(s,t)} + (n-k+t) \langle n-1 \rangle_{(s,t)} + \delta_{k0} \delta_{n0}, \quad (4.4)$$

with the additional conditions $\langle n \rangle_{(s,t)} = 0$ if $n < 0$ or $k < 0$. Comparing (4.4) with (1.17) (or (4.2)) with $q = 1$, we derive

$$\langle n \rangle_{(s,t)} = E_{n,k}(t, s, 1). \quad (4.5)$$

Remark 3. From (1.1) and (4.4) we derive

$$\langle n \rangle = \langle n \rangle_{(1,0)}, \quad \langle n+1 \rangle = \langle n \rangle_{(1,1)}. \quad (4.6)$$

Setting $q = 1$ in Eq. (1.23) we obtain

$$\frac{E_n(x | \alpha, \beta, 1)}{(1-x)^{n+\alpha+\beta}} = \sum_{j \geq 0} x^j (j+\beta)^n \binom{\alpha+\beta+j-1}{j}. \quad (4.7)$$

This is Proposition 12 in [2] by (4.5). Setting $q = 1$ in (1.24) we obtain

$$\sum_{n \geq 0} E_n(x | \alpha, \beta, 1) \frac{z^n}{n!} = \left(\frac{1-x}{1-xe^{(1-x)z}} \right)^\alpha \left(\frac{1-x}{e^{(x-1)z} - x} \right)^\beta, \quad (4.8)$$

which is equivalent to (1.5).

The r -Eulerian numbers, denoted by $A(n, k; r)$, count the number of permutations in $\mathfrak{S}_n$ that have exactly k r -descents (see (3.8)), that is,

$$A(n, k; r) := |\{\sigma \in \mathfrak{S}_n : \text{rdes}(\sigma) = k\}|. \quad (4.9)$$

From (3.13) with $q = 1$, we have

$$A(n+r, k; r) = (k+r)A(n+r-1, k; r) + (n+1-k)A(n+r-1, k-1; r) \quad (4.10)$$

for $0 \leq k \leq n$. Setting $(\alpha, \beta) = (r, 1)$ in (4.4) and combining with (4.10) and their initial value at $(n, k) = (0, 0)$, we derive

$$\langle n \rangle_{(r,1)} = \frac{1}{r!} A(n+r, k; r). \quad (4.11)$$

Remark 4. In view of (4.5) and (4.11) the 1-order (s, t) -Eulerian numbers coincide with the Carlitz-Scoville numbers and also encompass the r -Eulerian numbers. This fact was overlooked in [2, p.2], where it is stated that the Carlitz-Scoville numbers and the r -Eulerian numbers do not fall within their general framework of Eulerian numbers.

The identities (1.23), (1.25) and (3.14) deserve to have combinatorial interpretations. A combinatorial proof of (3.11) with $q = 1$ is given in [9]. Moreover, Butler [4] originally obtained the permutation interpretation of the q -Eulerian numbers $E_{n,k}(0, \beta, q)$ via the combinatorics of rook configurations (see also Haglund [12] and Garsia and Remmel [10]), it would be interesting to extend our results to rook configurations.

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